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Electric grasping apparatus for refuse vehicle

Abstract

A fully-electric grabber assembly includes a first grabber arm, a second grabber arm, an electric motor, and a plurality of gears. The plurality of gears includes a first gear coupled with the electric motor, an intermediate gear coupled with the first gear, and an arm gear coupled with the first grabber arm and the intermediate gear to facilitate pivoting the first grabber arm. The first gear is configured to rotate about an axis radially offset from a center thereof.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/885,311, filed Aug. 10, 2022, which is a continuation of U.S. patent application Ser. No. 16/851,162, filed Apr. 17, 2020, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/843,291, filed May 3, 2019, the entire disclosures of which are incorporated by reference herein.

BACKGROUND
(1) The present disclosure generally relates to the field of refuse vehicles, and in particular, to a grabber assembly for a refuse vehicle.
SUMMARY
(2) One implementation of the present disclosure is a fully-electric grabber assembly, according to an exemplary embodiment. The fully-electric grabber assembly includes a first grabber arm, a second grabber arm, an electric motor, and a plurality of gears. The plurality of gears includes a first gear coupled with the electric motor, an intermediate gear coupled with the first gear, and an arm gear coupled with the first grabber arm and the intermediate gear to facilitate pivoting the first grabber arm. The first gear is configured to rotate about an axis radially offset from a center thereof.
(3) Another implementation of the present disclosure is a fully-electric grabber assembly for a refuse vehicle, according to an exemplary embodiment. The fully-electric grabber assembly includes a carriage, a first grabber arm rotatably coupled with the carriage, a second grabber arm rotatably coupled with the carriage, an electric climb system configured to move the fully-electric grabber assembly relative to the refuse vehicle, and an electric gripping system. The electric gripping system includes an electric motor and a gear train coupling the electric motor to the first grabber arm to facilitate pivoting the first grabber arm. The gear train includes a first gear rotatably coupled with the carriage about an axis radially offset from a center of the first gear, and one or more second gears coupling the first gear to the first grabber arm. The first gear is driven by the electric motor.
(4) Another implementation of the present disclosure is a grabber assembly, according to an exemplary embodiment. The grabber assembly includes a grabber arm, an actuator, a first gear driven by the actuator, an intermediate gear coupled with the first gear, and an arm gear coupled with the grabber arm and the intermediate gear to facilitate pivoting the grabber arm. The first gear rotates about an axis radially offset from a center thereof.
(5) This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:
(2) FIG. **1** is a perspective view of a refuse vehicle, shown to include a loading assembly, a track, and a grabber assembly, according to an exemplary embodiment;
(3) FIG. **2** is a perspective view of the loading assembly of the refuse vehicle of FIG. **1**, according to an exemplary embodiment;
(4) FIG. **3** is a perspective view of the loading assembly of the refuse vehicle of FIG. **1**, shown to include the grabber assembly of FIG. **1**, according to an exemplary embodiment;
(5) FIG. **4** is a front view of the track of FIG. **1**, according to an exemplary embodiment;
(6) FIG. **5** is a side view of the track of FIG. **1**, according to an exemplary embodiment;
(7) FIG. **6** is a perspective view of the grabber assembly of FIG. **1**, shown to include a first grabber arm connector and a second grabber arm connector, according to an exemplary embodiment;
(8) FIG. **7** is a perspective view of the grabber assembly of FIG. **1**, shown to include a first grabber arm connector and a second grabber arm connector, according to an exemplary embodiment;
(9) FIG. **8** is an exploded view of the grabber assembly of FIG. **1**, according to an exemplary embodiment;
(10) FIG. **9** is a perspective view of the grabber assembly of FIG. **1** configured to be driven to grasp and release refuse containers by electric motors, according to an exemplary embodiment;
(11) FIG. **10** is a top view of the grabber assembly of FIG. **9** with an eccentric gearing system, according to an exemplary embodiment;
(12) FIG. **11** is a perspective view of an electric rack and pinion system for the grabber assembly of FIG. **1**, according to an exemplary embodiment;
(13) FIG. **12** is a perspective sectional view of an electric rack and pinion system for the grabber assembly of FIG. **1**, according to an exemplary embodiment;
(14) FIG. **13** is a perspective view of a lead screw and an electric motor configured to translate the lead screw, according to an

exemplary embodiment;
(15) FIG. **14** is a perspective view of an electrically actuated grabber assembly, according to an exemplary embodiment;
(16) FIG. **15** is a perspective view of the eccentric gearing system of the grabber assembly of FIG. **9**, according to an exemplary embodiment;
(17) FIGS. **16-19** are top views of the eccentric gearing system of the grabber assembly of FIG. **9** as a grabber arm is driven to swing, according to an exemplary embodiment;
(18) FIG. **20** is a perspective view of a grabber assembly, according to an exemplary embodiment;
(19) FIG. **21** is a perspective sectional view of the grabber assembly of FIG. **20**, according to an exemplary embodiment;
(20) FIG. **22** is a perspective sectional view of a portion of the grabber assembly of FIG. **20**, according to an exemplary embodiment; and
(21) FIG. **23** is a perspective view of the grabber assembly of FIG. **20** engaging a track of a refuse vehicle, according to an exemplary embodiment.

DETAILED DESCRIPTION

(22) Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.
(23) Referring generally to the FIGURES, a fully-electrically actuated grabber can include a carriage and grabber arms pivotally or rotatably coupled at opposite ends of the carriage. The carriage can be configured to translate along a track of a refuse vehicle. In other embodiments, the carriage is fixedly or removably coupled with an articulated arm, a telescoping arm, a boom, etc., of the refuse vehicle. The fully-electrically actuated grabber can also be implemented in a variety of systems or devices other than a refuse vehicle, including, for example, a telehandler, a boom lift, a front loading refuse vehicle, a rear loading refuse vehicle, a side loading refuse vehicle, etc.
(24) The fully-electrically actuated grabber can use a variety of electrically activated systems. For example, the fully-electrically actuated grabber can include any number of motors, electric linear actuators, etc. The fully-electrically actuated grabber can include a fully electric rack and pinion system that uses an electric motor to produce side-to-side translation of rack members. The rack members can include teeth that mesh with gearing systems to pivot/rotate the grabber arms as the rack members translate. In other embodiments, an electric motor directly drives a gearing system configured to pivot/rotate one or both of the grabber arms. The gearing system may include any number of gears, a gear train, etc. In some embodiments, the gearing system is an eccentric gearing system.
(25) In other embodiments, electric linear actuators are used to pivot/rotate each of the grabber arms. The electric linear actuators can be pivotally coupled at one end with the carriage, and at a distal end with one of the grabber arms. Extension/expansion and retraction/compression of the electric linear actuators can independently pivot/rotate each of the grabber arms. The fully-electrically actuated grabber can also use an eccentric gearing system. Various embodiments described herein are configured to independently drive the grabber arms to swing or pivot, while other embodiments drive the grabber arms to swing or pivot concurrently through a unified system (e.g., a system that drives the grabber arms to swing or pivot simultaneously). s
(26) According to the exemplary embodiment shown in FIG. **1**, a vehicle, shown as refuse vehicle **10** (e.g., a garbage truck, a waste collection truck, a sanitation truck, a refuse collection truck, a refuse collection vehicle, etc.), is configured as a side-loading refuse truck having a first lift mechanism/system (e.g., a side-loading lift assembly, etc.), shown as lift assembly **100**. In other embodiments, refuse vehicle **10** is configured as a front-loading refuse truck or a rear-loading refuse truck. In still other embodiments, the vehicle is another type of vehicle (e.g., a skid-loader, a telehandler, a plow truck, a boom lift, etc.).
(27) As shown in FIG. **1**, refuse vehicle **10** includes a chassis, shown as frame **12**; a body assembly, shown as body **14**, coupled to frame **12** (e.g., at a rear end thereof, etc.); and a cab, shown as cab **16**, coupled to frame **12** (e.g., at a front end thereof, etc.). Cab **16** may include various components to facilitate operation of refuse vehicle **10** by an operator (e.g., a seat, a steering wheel, hydraulic controls, a user interface, switches, buttons, dials, etc.). As shown in FIG. **1**, refuse vehicle **10** includes a prime mover, shown as engine **18**, coupled to frame **12** at a position beneath cab **16**. Engine **18** is configured to provide power to a plurality of tractive elements, shown as wheels **19**, and/or to other systems of refuse vehicle **10** (e.g., a pneumatic system, a hydraulic system, an electric system, etc.). Engine **18** may be configured to utilize one or more of a variety of fuels (e.g., gasoline, diesel, bio-diesel, ethanol, natural gas, etc.), according to various exemplary embodiments. According to an alternative embodiment, engine **18** additionally or alternatively includes one or more electric motors coupled to frame **12** (e.g., a hybrid refuse vehicle, an electric refuse vehicle, etc.). The electric motors may consume electrical power from an on-board storage device (e.g., batteries, ultra-capacitors, etc.), from an on-board generator (e.g., an internal combustion engine, etc.), and/or from an external power source (e.g., overhead power lines, etc.) and provide power to the systems of refuse vehicle **10**.
(28) According to an exemplary embodiment, refuse vehicle **10** is configured to transport refuse from various waste receptacles within a municipality to a storage and/or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). As shown in FIG. **1**, body **14** includes a plurality of panels, shown as panels **32**, a tailgate **34**, and a cover **36**. Panels **32**, tailgate **34**, and cover **36** define a collection chamber (e.g., hopper, etc.), shown as refuse compartment **30**. Loose refuse may be placed into refuse compartment **30** where it may thereafter be compacted. Refuse compartment **30** may provide temporary storage for refuse during transport to a waste disposal site and/or a recycling facility. In some embodiments, at least a portion of body **14** and refuse compartment **30** extend in front of cab **16**. According to the embodiment shown in FIG. **1**, body **14** and refuse compartment **30** are positioned behind cab **16**. In some embodiments, refuse compartment **30** includes a hopper volume and a storage volume. Refuse may be initially loaded into the hopper volume and thereafter compacted into the storage volume. According to an exemplary embodiment, the hopper volume is positioned between the storage volume and cab **16** (i.e., refuse is loaded into a position of refuse compartment **30** behind cab **16** and stored in a position further toward the rear of refuse compartment **30**). In other embodiments, the storage volume is positioned between the hopper volume and cab **16** (e.g., a rear-loading refuse vehicle, etc.).
(29) As shown in FIG. **1**, refuse vehicle **10** includes first lift mechanism/system (e.g., a front-loading lift assembly, etc.), shown as lift assembly **100**. Lift assembly **100** includes a grabber assembly, shown as grabber assembly **42**, movably coupled to a track,

shown as track **20**, and configured to move along an entire length of track **20**. According to the exemplary embodiment shown in FIG. **1**, track **20** extends along substantially an entire height of body **14** and is configured to cause grabber assembly **42** to tilt near an upper height of body **14**. In other embodiments, track **20** extends along substantially an entire height of body **14** on a rear side of body **14**. Refuse vehicle **10** can also include a reach system or assembly coupled with a body or frame of refuse vehicle **10** and lift assembly **100**. The reach system can include telescoping members, a scissors stack, etc., or any other configuration that can extend or retract to provide additional reach of grabber assembly **42** for refuse collection.

(30) Referring still to FIG. **1**, grabber assembly **42** includes a pair of grabber arms shown as grabber arms **44**. Grabber arms **44** are configured to rotate about an axis extending through a bushing. Grabber arms **44** are configured to releasably secure a refuse container to grabber assembly **42**, according to an exemplary embodiment. Grabber arms **44** rotate about the axis extending through the bushing to transition between an engaged state (e.g., a fully grasped configuration, a fully grasped state, a partially grasped configuration, a partially grasped state) and a disengaged state (e.g., a fully open state/configuration, a fully released state/configuration, a partially open state/configuration, a partially released state/configuration). In the engaged state, grabber arms **44** are rotated towards each other such that the refuse container is grasped therebetween. In the disengaged state, grabber arms **44** rotate outwards (as shown in FIG. **3**) such that the refuse container is not grasped therebetween. By transitioning between the engaged state and the disengaged state, grabber assembly **42** releasably couples the refuse container with grabber assembly **42**. Refuse vehicle **10** may pull up along-side the refuse container, such that the refuse container is positioned to be grasped by the grabber assembly **42** therebetween. Grabber assembly **42** may then transition into an engaged state to grasp the refuse container. After the refuse container has been securely grasped, grabber assembly **42** may be transported along track **20** with the refuse container. When grabber assembly **42** reaches the end of track **20**, grabber assembly **42** may tilt and empty the contents of the refuse container in refuse compartment **30**. The tilting is facilitated by the path of track **20**. When the contents of the refuse container have been emptied into refuse compartment **30**, grabber assembly **42** may descend along track **20**, and return the refuse container to the ground. Once the refuse container has been placed on the ground, the grabber assembly may transition into the disengaged state, releasing the refuse container.

(31) Referring now to FIGS. **2-3**, the lift assembly **100** is shown in greater detail, according to an exemplary embodiment. Lift assembly **100** is shown to include track **20**, and a coupling member, shown as connecting member **26**. Track **20** is configured to extend along substantially the entire height of body **14**, according to the exemplary embodiment shown. Body **14** is shown to include a loading section, shown as loading section **22**. Loading section **22** is shown to include a recessed portion, shown as recessed portion **24**. Recessed portion **24** is configured to allow track **20** to curve through recessed portion **24**, such that track **20** may be configured to empty a refuse bin (e.g., a garbage can) releasably couple to grabber assembly **42** in refuse compartment **30**.

(32) Still referring to FIGS. **2-3**, connecting member **26** is shown coupled with track **20**. Connecting member **26** is coupled to track **20** such that connecting member **26** may move along an entire path length of track **20**. Connecting member **26** may removably couple with grabber assembly **42**, thereby removably coupling grabber assembly **42** to track **20**, and allowing grabber assembly **42** to travel along the entire path length of track **20**. Connecting member **26** removably couples (e.g., by removable fasteners) to a carriage portion of grabber assembly **42**, shown as carriage **46**. Grabber assembly **42** is shown to include grabber arms, shown as first grabber arm **44***a* and second grabber arm **44***b*, according to an exemplary embodiment. First grabber arm **44***a* and second grabber arm **44***b* are each configured to pivot about axis **45***a* and axis **45***b*, respectively. Axis **45***a* is defined as an axis longitudinally extending through substantially an entire length of a first adapter or bushing assembly, shown as first adapter assembly **43***a*, and axis **45***b* is defined as an axis longitudinally extending through substantially an entire length of a second adapter or bushing assembly, shown as second adapter assembly **43***a*. First adapter assembly **43***a* fixedly couples to a first end of carriage **46**, and rotatably couples to first grabber arm **44***a*. Second adapter assembly **43***b* fixedly couples to a second end of carriage **46**, and rotatably couples to second grabber arm **44***b*. First adapter assembly **43***a* and second adapter assembly **43***b* couple first grabber arm **44***a* and second grabber arm **44***b* to carriage **46**, and allow first grabber arm **44***a* and second grabber arm **44***b* to rotate about axis **45***a* and axis **45***b*, respectively.

(33) Referring now to FIGS. **4-5**, the track **20** is shown in greater detail according to an exemplary embodiment. FIG. **4** shows a front view of track **20**, and FIG. **5** shows a side view of track **20**, according to an exemplary embodiment. Track **20** is shown to include a straight portion **27**, and a curved portion **29**. Straight portion **27** may be substantially vertical, and/or substantially parallel to loading section **22** of body **14**, according to an exemplary embodiment. Curved portion **29** may have a radius of curvature, shown as radius **23**, according to an exemplary embodiment. In some embodiments, curved portion **29** has a constant radius of curvature (i.e., curved portion **29** has a constant radius **23** along all points on a path of curved portion **29**), while in other embodiments, curved portion **29** has a non-constant radius of curvature (i.e., curved portion **29** has a non-constant radius **23** along various points on the path of curved portion **29**). According to an exemplary embodiment, straight portion **27** has an infinite radius of curvature. According to an exemplary embodiment, grabber assembly **42** may travel along a path of track **20**, shown as path **25**. Track **20** may be configured to tilt grabber assembly **42** to empty contents of a refuse container when grabber assembly **42** travels along path **25** and travels past a point on path **25**, shown as point **28**. When grabber assembly **42** travels along path **25** past point **28**, grabber assembly **42** may tilt, emptying the contents of the refuse container in refuse compartment **30**.

(34) Referring now to FIGS. **6-7**, grabber assembly **42** is shown, according to an exemplary embodiment. Grabber assembly **42** includes carriage **46**, first adapter assembly **43***a* fixedly coupled at the first end of carriage **46**, second adapter assembly **43***a* fixedly coupled at the second end of carriage **46**, first grabber arm **44***a* rotatably coupled to first adapter assembly **43***a*, and second grabber arm **44***b* rotatably coupled to second adapter assembly **43***a*. Grabber assembly **42** also includes hooks, shown as hooks **48**, according to an exemplary embodiment. Hooks **48** may be integrally formed with carriage **46**, according to some embodiments. In some embodiments, hooks **48** are removably coupled to carriage **46** (e.g., with fasteners). Hooks **48** may assist in removably coupling grabber assembly **42** to connecting member **26**, according to an exemplary embodiment. Hooks **48** may aid in supporting the weight of grabber assembly **42** when grabber assembly **42** is attached to connecting member **26**, according to some embodiments.

(35) Referring still to FIGS. **6-7**, carriage **46** is shown to have a profile that is the shape of a portion of a hexagon. In other embodiments, carriage **46** has a curved profile, a straight profile, a rectangular profile, an irregularly shaped profile, etc. The

profile of carriage **46** may assist in coupling grabber assembly **42** to a refuse container by interfacing with a curved or correspondingly shaped portion of the refuse container, according to an exemplary embodiment. Referring still to FIGS. **6** and **7**, each of first grabber arm **44***a*, and second grabber arm **44***b* are shown to include a curved profile, according to an exemplary embodiment. First grabber arm **44***a* and second grabber arm **44***b* can have a generally arcuate shape that can facilitate surrounding and grasping refuse collection bins.
(36) Referring still to FIGS. **6-7**, grabber assembly **42** is shown to include first grabber arm **44***a* and second grabber arm **44***b*, according to an exemplary embodiment. Second grabber arm **44***b* includes a first protrusion, shown as first grabber finger **50***a*, and a second protrusion shown as second grabber finger **50***b*, according to an exemplary embodiment. First grabber finger **50***a* and second grabber finger **50***b* define an open space, shown as open space **52**. Open space **52** may have a width equal to or greater than a maximum width of first grabber arm **44***a*, according to some embodiments. In some embodiments, when grabber assembly **42** transitions into an engaged state (e.g., a grasped state, a fully grasped state, etc.), first grabber arm **44***a* moves into the open space **52**, defined by the space between first grabber finger **50***a* and second grabber finger **50***b*.
(37) Referring now to FIG. **8**, an exploded view of a portion of grabber assembly **42** is shown in greater detail, according to an illustrative embodiment. First adapter assembly **43***a* and second adapter assembly **43***b* are symmetrical, such that whatever is said of first adapter assembly **43***a* may be said of second adapter assembly **43***b*, and vice versa. First adapter assembly **43***a* is shown removably disconnected from carriage **46**, according to the illustrative embodiment. First adapter assembly **43***a* includes a top piece, shown as top adapter piece **56***a*, a bottom piece, shown as bottom adapter piece **58***a* and a pin, shown as adapter assembly pin **60***a*. Top adapter piece **56***a* coupled with a plate, shown as top adapter plate **62***a*, according to an exemplary embodiment. In some embodiments, top adapter plate **62***a* is integrally formed with top adapter piece **56***a*. Top adapter plate **62***a* is shown to include fasteners **66***a*, according to an exemplary embodiment. Top adapter plate **62***a* is configured to couple with a surface, shown as surface **74***a* of carriage **46**, and removably couples to carriage **46** via fasteners **66***a* and fastener connections **76***a*. In some embodiments, fasteners **66***a* include a bolt and a nut, and fastener connections **76***a* are through holes configured to receive fasteners **66***a* to removably couple first adapter assembly **43***a* to carriage **46**. First adapter assembly **43***a* is also shown to include bottom adapter piece **58***a*, according to an exemplary embodiment. Bottom adapter piece **58***a* may be constructed similarly to top adapter piece **56***a*, and includes a bottom plate, shown as bottom adapter plate **64***a*. Bottom adapter plate **64***a* includes bottom fasteners, shown as fasteners **68***a*, and configured to removably couple first adapter assembly **43***a* to carriage **46**. Fasteners **68***a* are shown to be the same as fasteners **66***a*, and may be configured couple with a bottom surface of carriage **46**, vis a vis fasteners **66***a*.
(38) Referring still to FIG. **8**, first adapter assembly **43***a* includes adapter assembly pin **60***a*. Adapter assembly pin **60***a* is configured to pivotally couple with a bushing, shown as first bushing **54***a* of first grabber arm **44***a*, according to an exemplary embodiment. Adapter assembly pin **60***a* may allow first grabber arm **44***a* to pivot about axis **45***a* and support first grabber arm **44***a*. Adapter assembly pin **60***a* is configured to extend through an entire longitudinal length of first bushing **54***a* and extends into top adapter piece **56***a* at a first end, and into bottom adapter piece **58***a* at a second end, according to an exemplary embodiment. First adapter assembly **43***a* is also shown to include a top fastener and a bottom fastener, shown as top pin fastener **70***a* and bottom pin fastener **72***a*, according to an exemplary embodiment. Top pin fastener **70***a* and bottom pin fastener **72***a* removably couple adapter assembly pin **60***a* to the top adapter piece **56***a* and the bottom adapter piece **58***a*, respectively. In an exemplary embodiment, top pin fastener **70***a* and bottom pin fastener **72***a* extend through axis **45***a*. Axis **45***a* may be defined as an axis extending longitudinally through a center of adapter assembly pin **60***a*. Top adapter piece **56***a* and bottom adapter piece **58***a* may be axially aligned with adapter assembly pin **60***a*, such that axis **45***a* extends through the center of top adapter piece **56***a* and bottom adapter piece **58***a*, according to an exemplary embodiment. In some embodiments, each of top pin fastener **70***a* and bottom pin fastener **72***a* are bolt fasteners and extend through a top hole and a bottom hole, respectively. The top hole is normal to axis **45***a*, and radially extends through top adapter piece **56***a* and the end of adapter assembly pin **60***a* which extends into top adapter piece **56***a*, according to an exemplary embodiment. The bottom hole is normal to axis **45***a* and radially extends through bottom adapter piece **58***a* and the end of adapter assembly pin **60***a* which extends into bottom adapter piece **58***a*, according to an exemplary embodiment.
(39) Referring still to FIG. **8**, first grabber arm **44***a* and second grabber arm **44***b* are shown to include first bushing **54***a* and second bushing **54***b*, according to an exemplary embodiment. First bushing **54***a* and second bushing **54***b* are configured to pivotally couple with adapter assembly pin **60***a* and adapter assembly pin **60***b*, respectively. Each of first bushing **54***a* and second bushing **54***b* include a connecting feature, shown as connecting feature **82***a* and connecting feature **82***b*, respectively. First bushing **54***a* and second bushing **54***b* may be similarly or symmetrically configured and constructed, such that whatever is said of first bushing **54***a* may be said of second bushing **54***b*, and vice versa. First bushing **54***a* removably couples to first grabber arm **44***a* through connecting feature **82***a*, according to an exemplary embodiment. First bushing **54***a* may include a flanged portion (not shown) which extends behind a plate portion **84***a* of first grabber arm **44***a*. Connecting feature **82***a* may be positioned on an opposite side of plate portion **84***a* of first grabber arm **44***a* and may include fasteners configured to removably couple first bushing **54***a* with plate portion **84***a* of first grabber arm **44***a*.
(40) Referring still to FIG. **8**, each of first adapter assembly **43***a* and second adapter assembly **43***b* are shown removably disconnected from carriage **46**. Each of top adapter piece **56***a* and bottom adapter piece **58***a* are shown removably coupled to adapter assembly pin **60***a*. First bushing **54***a* and second bushing **54***b* are shown removably coupled to first grabber arm **44***a* and second grabber arm **44***b*, respectively. The removable couplings discussed in greater detail above are advantageous since they allow grabber assembly **42** to be quickly and easily disassembled for maintenance purposes. A technician may easily disassemble grabber assembly **42** by first disconnecting top fasteners **66***a*, bottom fasteners **68***a*, top fasteners **66***b*, and bottom fasteners **68***b* to remove first adapter assembly **43***a* and second adapter assembly **43***b*. The technician may further disassemble first adapter assembly **43***a* and second adapter assembly **43***b* by disconnecting top pin fastener **70***a*, bottom pin fastener **72***a*, top pin fastener **70***b*, and bottom pin fastener **72***b* to completely disassemble first adapter assembly **43***a* and second adapter assembly **43***b*. Additionally, the technician may disassembly first bushing **54***a* and second bushing **54***b* by disconnecting connecting feature **82***a* and connecting feature **82***b* from first grabber arm **44***a* and second grabber arm **44***b*. This allows the technician to quickly disassemble and replace various components of grabber assembly **42** and may decrease an amount of maintenance time required for the refuse vehicle **10**, allowing refuse vehicle **10** to quickly return to service. In some embodiments, the removable couplings

described also allow the technician to adjust an alignment of the adapter assembly pin (e.g., adapter assembly pin **60***a*) within the bushing (e.g., first bushing **54***a*). For example, if top fasteners **66***a* and bottom fasteners **68***a* are bolt fasteners, fastener connections **76***a* may be through holes, configured to receive fasteners **66***a*. The diameters of fastener connections **76***a* may be large enough to allow fasteners **66***a* to be adjusted within the diameter of fastener connections **76***a*, to adjust the alignment of adapter assembly pin **60***a* within bushing **54***a*. The technician may loosen top fasteners **66***a* or bottom fasteners **68***a* and adjust a position of top fasteners **66***a* or bottom fasteners **68***a* within fastener connections **76***a*. Once a desired alignment has been achieved, the technician may tighten top fasteners **66***a* and/or bottom fasteners **68***a* to maintain the desired alignment.
(41) In some embodiments, grabber assembly **42** uses polymeric wear surfaces. For example, an inner diameter of first bushing **54***a* may undergo wear due to the slidable coupling between the inner diameter of first bushing **54***a* and an outer diameter of adapter assembly pin **60***a*. In order to provide proper lubrication and/or wear resistance, a polymeric surface may be applied to at least one of the outer diameter of adapter assembly pin **60***a* and the inner diameter of first bushing **54***a*, according to an exemplary embodiment. The polymeric wear surface may be any of a surface made of polyethylene, polytetrafluoroethylene, polypropylene, polyisobutylene, polystyrene, polyvinylchloride, polyehterketone, polyoxymethylene, polyimide, etc., or any other polymer. In some embodiments, the polymeric wear surface may be a sleeve removably inserted into the first bushing **54***a* or over the adapter assembly pin **60***a*. Advantageously, polymeric wear surfaces reduce the need to periodically grease the wear surfaces of grabber assembly **42**. This may result in cost and time savings by providing a greaseless grabber assembly **42**.
(42) Referring now to FIGS. **9** and **10**, first grabber arm **44***a* and second grabber arm **44***b* can be driven to grasp (e.g., to pivot or rotate about adapter assemblies **43**) by one or more electric motors **154**. Specifically, second grabber arm **44***b* can be driven to rotate inwards or outwards (e.g., as shown in FIG. **8**) by electric motor **154***b* and first grabber arm **44***a* can be driven to rotate inwards or outwards (e.g., as shown in FIG. **8**) by electric motor **154***a*. First grabber arm **44***a* can rotate about axis **45***a* in a clockwise direction to grasp a refuse container and in a counter-clockwise direction about axis **45***a* to release a refuse container. Likewise, second grabber arm **44***b* can rotate about axis **45***b* in a counter-clockwise direction about axis **45***b* to grasp a refuse container and in a clockwise direction about axis **45***b* to release the refuse container. Electric motors **154** can be positioned at opposite ends of carriage **46**.
(43) In some embodiments, first grabber arm **44***a* and second grabber arm **44***b* are configured to be driven to pivot or rotate about axes **45** independently. In other embodiments, the rotation (e.g., degree and/or direction) of first grabber arm **44***a* and second grabber arm **44***b* are related (e.g., through a gearing system, with connection members, etc.).
(44) Referring particularly to FIGS. **10** and **15**, grabber assembly **42** can include a gearing system **200** to transfer rotational kinetic energy from one of electric motors **154** to a corresponding or associated one of grabber arms **44**. For example, gearing system **200** as shown in FIG. **9** is configured to transfer rotational kinetic energy from electric motor **154***b* to second grabber arm **44***b*. A similar and/or symmetric gearing system can be used at an opposite or distal end of carriage **46** to transfer rotational kinetic energy from electric motor **154***a* to first grabber arm **42**.
(45) Gearing system **200** is positioned at an end of carriage **46** and is configured to transfer the rotational kinetic energy output by a driveshaft of electric motor **154***b* to second grabber arm **44***b* to pivot/rotate second grabber arm **44***b* about axis **45***b*. Gearing system **200** may be an eccentric gearing system. Gearing system **200** includes an input gear **202**, a driven gear **206**, and an output gear **210**. Input gear **202** is configured to receive rotational kinetic energy from the driveshaft of electric motor **154***b*. Input gear **202** includes teeth **204** along an entire outer periphery or an outer surface. Teeth **56** of input gear **202** are configured to mesh with teeth **208** of driven gear **206**. In some embodiments, input gear **202**, driven gear **206**, and output gear **210** are spur gears. In other embodiments, input gear **202**, driven gear **206**, and output gear **210** are helical gears. It should be understood that input gear **202**, driven gear **206**, and output gear can be any types of gears. A similar and/or symmetric (e.g., mirrored) gearing system can be positioned at an opposite end of carriage **46** to transfer the rotational kinetic energy output by a driveshaft of electric motor **154***a* to first grabber arm **44***a* to pivot/rotate first grabber arm **44***a* about axis **45***a*.
(46) Output gear **210** includes teeth **212** along only a portion of an outer periphery or an outer surface thereof. In some embodiments, output gear **210** includes teeth **212** along an entire perimeter or outer periphery/surface thereof. Teeth **212** of output gear **210** is configured to mesh with teeth **208** of driven gear **206**. Output gear **210** can have the form of a sector of a circle, with teeth **212** disposed or formed along an outer radius thereof. In some embodiments, output gear **210** is configured to pivot an angular amount that is less than 360 degrees.
(47) Input gear **202** is rotatably coupled with carriage **46** near the end of carriage **46** that rotatably/pivotally couples with second grabber arm **44***b*. Input gear **202** can be fixedly or rotatably coupled with a shaft that fixedly or rotatably couples with carriage **46**. For example, input gear **202** can be rotatably mounted (e.g., with a bearing) to the shaft that fixedly couples with carriage **46**. In other embodiments, input gear **202** is fixedly coupled with the shaft that rotatably couples (e.g., with a bearing) with carriage **46**. In either case, input gear **202** is translationally fixedly coupled with carriage **46**, but can rotate relative to carriage **46**. In some embodiments, input gear **202** is rotatably coupled with carriage **46** at an aperture, a hole, a bore, a recess, etc., shown as aperture **60**. Aperture **60** is radially offset from a central axis **216** of input gear **202**. In some embodiments, a shaft extends therethrough aperture **60** and rotatably couples input gear **202** with carriage **46** such that input gear **202** rotates about axis **220** that extends therethrough a center of aperture **60**.
(48) In some embodiments, input gear **202** is driven to rotate by electric motor **154***b* through aperture **60**. Aperture **60** can extend at least partially (or completely) through a thickness of input gear **202** and is configured to receive a pin, a cylindrical member, a post, etc., therethrough. In some embodiments, electric motor **154***b* is configured to drive input gear **202** with a post, a pin, a cylindrical member, a correspondingly shaped protrusion, etc., that is a driveshaft of electric motor **154***b* and is received therewithin aperture **60**. In some embodiments, the post that is received within aperture **60** is slidably or fixedly coupled with an inner surface, an inner periphery, an inner perimeter, etc., of aperture **60**. In some embodiments, electric motor **154***b* drives input gear **202** to rotate about axis **220** through a coupling therebetween the driveshaft and a protrusion **222**. Protrusion **222** is positioned at a center point of input gear **202**.
(49) Input gear **202** is configured to drive gear **206** through the meshed coupling therebetween. In some embodiments, gear **206** is rotatably coupled with carriage **46**. In some embodiments, a linkage, a coupling member, a bar, a beam, etc., shown as linkage **226**

(see FIG. **19**). For example, linkage **226** can extend therebetween center points of input gear **202** and gear **206** to rotatably couple input gear **202** and gear **206**. In some embodiments, gear **206** includes a protrusion **224** similar to protrusion **222**. Protrusion **224** is positioned at a center point of gear **206**. In some embodiments, linkage **226** extends therebetween protrusion **222** and protrusion **224** to link input gear **202** and gear **206**. Linkage **226** can include apertures, bores, holes, etc., at opposite ends that are correspondingly shaped with protrusions **222** and **224**. In some embodiments, protrusions **222** and **224** are received therewithin the correspondingly shaped apertures of linkage **226** and slidably or pivotally couple with the correspondingly shaped apertures. In this way, a distance therebetween axis **216** and axis **218** (axis **218** extends through a center point of gear **206**) is fixed. Gear **206** can rotate or pivot about axis **218**. Additionally, gear **206** can rotate or pivot about axis **220** of input gear **202** as input gear **202** is driven to rotate about axis **220** by electric motor **154***b*.

(50) Gear **206** is configured to drive output gear **210**, according to an exemplary embodiment. Output gear **210** can be fixedly coupled with an elongated member, a cylindrical member, a post, etc., of second grabber arm **44***b*. In some embodiments, output gear **210** is fixedly coupled with a portion of adapter assembly **43***b* that pivots/rotates with second grabber arm **44***b*. In some embodiments, an inner periphery of a central aperture of output gear **210** is fixedly coupled with an outer periphery of a cylindrical member of adapter assembly **43***b*. Output gear **210** can be mounted on (e.g., fixedly coupled with) bushing **54***b*. In some embodiments, output gear **210** is press fit or interference fit with an outer perimeter or an outer periphery of bushing **54***b*. Output gear **210** can be positioned at any position along a height of bushing **54***b*. In some embodiments, output gear **210** is mounted to bushing **54***b* at a bottom end, a top end, or at a position therebetween.

(51) A linkage **214** extends therebetween output gear **210** and gear **206**. Linkage **214** can be the same as or similar to linkage **226**. Linkage **214** can include apertures at opposite ends that rotatably or slidably couple with protrusion **224** and bushing **54***b*. In some embodiments, a distance between axis **218** and axis **45***b* is fixed due to linkage **214**. However, gear **206** can rotate about axis **45***b* relative to output gear **210**. Linkage **214** is free to rotate or pivot about axis **45***b*. Linkage **214** can be driven to rotate about axis **45***b* due to rotation of input gear **202** about axis **220** and rotation and translation of gear **206** relative to input gear **202**.

(52) Output gear **210** can be driven to rotate/pivot about axis **45***b*. In some embodiments, axis **45***b* extends through the center point of output gear **210**. Output gear **210** is configured to rotate at least an angular amount θ.sub.range. In some embodiments, the angular amount θ.sub.range is an angular amount between a fully released position of second grabber arm **44***b* (shown in FIG. **16**) and a fully grasped position of second grabber arm **44***b*. For example, second grabber arm **44***b* can be configured to rotate/pivot an angular amount θ.sub.range=90°, an angular amount θ.sub.range=120°, an angular amount θ.sub.range=160°, etc., or any other angular amount. In some embodiments, the angular amount θ.sub.range is a maximum allowable angular amount that each of grabber arms **44** can rotate from the fully released position to a position/configuration where end portions of grabber arms **44** are substantially in contact.

(53) Gearing system **200** can be positioned inside of carriage **46**. For example, the various gears, shafts, etc., of gearing system **200** can be mounted or rotatably coupled with an inner surface of a bottom member of carriage **46**. Carriage **46** can be a hollow structural member having top and bottom structural members (e.g., plates) that are substantially parallel to each other and define an overall height of carriage **46**. Gearing system **200** can be positioned at an inner surface of either of the top and bottom structural members. For example, gearing system **200** can be positioned at a top (an interior) surface of the bottom structural member or at a bottom (an interior) surface of the top structural member. In other embodiments, gearing system **200** is positioned outside of carriage **46** (e.g., on a top surface or on a bottom surface of carriage **46**). If gearing system **200** is positioned outside of an inner volume of carriage **46**, a housing member can be removably coupled with carriage **46** to contain at least a portion of gearing system **200** therewithin.

(54) Output gear **210** can be driven to rotate in a first direction (e.g., clockwise) about axis **45***b* to pivot/rotate second grabber arm **44***b* into the grasped configuration. Likewise, output gear **210** can be driven to rotate in a second direction (e.g., counter-clockwise) about axis **45***b* to pivot/rotate second grabber arm **44***b* into the fully released configuration. In some embodiments, electric motor **154***b* is a reversible motor and can be driven in either direction. In this way, electric motor **154***b* can be operated (e.g., by a control system) to drive output gear **210** to rotate/pivot in either direction, thereby driving second grabber arm **44***b* to pivot about axis **45***b*. The torque output by electric motor **154***b* can be transferred through gearing system **200** to second grabber arm **44***b* to rotate/pivot second grabber arm **44***b* relative to carriage **46** about axis **45***b*.

(55) Referring particularly to FIGS. **16-19**, the operation of gearing system **200** is shown in greater detail. Gearing system **200** can be driven by electric motor **154***b* to pivot second grabber arm **44***b* about axis **45***b*. FIG. **16** shows the configuration of gearing system **200** when second grabber arm **44***b* is in a fully released configuration. FIGS. **17-18** show the configuration of gearing system **200** as input gear **202** is driven to rotate about axis **220** that extends therethrough aperture **60**.

(56) As gearing system **200** is driven by electric motor **154***b*, input gear **202** is driven to rotate about axis **220** (e.g., in a clockwise direction as shown in FIGS. **16-19**). The center point of input gear **202** can rotate about axis **220** in the clockwise direction. This results in gear **206** being driven to rotate about axis **220** or central axis **216** relative to input gear **202** in a counter clockwise direction. Due to the meshed (e.g., through the teeth) coupling therebetween input gear **202** and gear **206**, gear **206** also rotates about central axis **218**. As gear **206** rotates about central axis **218** and swings about central axis **216** or axis **220** relative to input gear **202**, output gear **210** is driven to rotate in the clockwise direction due to the meshed coupling (through the teeth) therebetween gear **206** and output gear **210**. Rotation of output gear **210** in the clockwise direction results in second grabber arm **44***b* being rotated about axis **45***b* in the clockwise direction relative to carriage **46**.

(57) In this way, electric motor **154***b* can be configured to drive input gear **202** to rotate about axis **220** in the clockwise direction to pivot/rotate second grabber arm **44***b* about axis **45***b* in the clockwise direction (e.g., to grasp containers). Electric motor **154***b* can also drive input gear **202** to rotate about axis **220** in the counter clockwise direction, thereby pivoting/rotating second grabber arm **44***b* about axis **45***b* in the counter clockwise direction (e.g., to release containers). It should be understood that first grabber arm **44***a* can include a gearing system that is the same as or similar to (e.g., symmetric) gearing system **200**.

(58) Using an eccentric gearing system as shown in FIGS. **9-10** and **15-19** facilitates improved clamping force therebetween first grabber arm **44***a* and second grabber arm **44***b* to grasp containers. Specifically, an initial angular speed of first grabber arm **44***a* and second grabber arm **44***b* when transitioning from the fully release configuration to a partially grasped configuration is relatively

high and the exerted torque is relatively low. However, as first grabber arm **44***a* and second grabber arm **44***b* swing to the fully grasped or to a more grasped configuration (e.g., as second grabber arm **44***b* swings about axis **45***b* in the clockwise direction as shown in FIGS. **16-19**), the angular speed of grabber arms **44** decreases while the torque exerted increases. This facilitates increased clamping force as first and second grabber arms **44** transition or are driven to rotate into a more grasped configuration.
(59) Gearing system **200** converts a constant rotational speed input (e.g., at input gear **202**) to a variable output rotational arm speed. For example, input gear **202** can be rotated by electric motor **154***b* at a constant angular speed that results in a varying angular speed of second grabber arm **44***b* as second grabber arm **44***b* rotates about axis **45***b*. Gearing system **200** provides a variable gear ratio as second grabber arm **44***b* swings about axis **45***b*. This facilitates a faster rotational speed of second grabber arm **44***b* under no-load as well as improved clamping force and precision when second grabber arm **44***b* approaches the fully-clamped configuration.
(60) Referring now to FIG. **11**, an electric rack and pinion system **300** can be used to drive first grabber arm **44***a* and second grabber arm **44***b*, according to an exemplary embodiment. Electric rack and pinion system **300** includes an electric motor **354** configured to drive rack members **310** and **309**. Electric motor **354** is configured to drive rack members **310** and **309** to translate in opposing directions.
(61) Electric motor **354** can include an output driveshaft and an output gear **318** to transfer rotational kinetic energy to translate rack members **310** and **309**. Electric motor **354** can include a gear box configured to increase torque output provided at output gear **318**. Output gear **318** is configured to mesh with teeth **316** of rack member **309**. Rack member **309** includes a first member **304** and a second member **306**. Member **304** and member **306** can be integrally formed with each other. Member **304** includes teeth **316** configured to mesh with teeth of output gear **318**. Member **306** includes teeth **308** configured to mesh with teeth of arm gear **302***a*. Teeth **316** and teeth **308** can be positioned on opposite surfaces of rack member **309**.
(62) Arm gear **302***a* can be integrally formed or fixedly coupled with bushing **54***a* of first grabber arm **44***a*. In some embodiments, arm gear **302***a* is positioned at a bottom end of bushing **54***a*. In other embodiments, arm gear **302***a* is positioned at an upper or top end of bushing **54***a*. In still other embodiments, arm gear **302***a* is positioned between the upper or top end and the bottom end of bushing **54***a*. In some embodiments, arm gear **302***a* is press fit on an outer surface, an outer perimeter, an outer periphery, etc., of bushing **54***a*.
(63) Rack member **310** can include a first portion **312** and a second portion **314**. First portion **312** and second portion **314** can be integrally formed with each other. First portion **312** and second portion **314** can be elongated structural members, tubular members, square tubular members, square structural members, rectangular members, circular tubular members, etc., or any other structural member that provides sufficient length and can provide sufficient tensile and compressive strength. For example, rack member **310** can be a steel member.
(64) First portion **312** includes teeth on a first surface configured to mesh with teeth of output gear **318**. Second portion **314** includes teeth **316** on a surface configured to mesh with teeth of arm gear **302***b*. In some embodiments, the surface of first portion **312** that includes the teeth configured to mesh with teeth of output gear **318** is substantially parallel with the surface of second portion **314** that includes teeth **316**.
(65) Second portion **314** and first portion **312** can be laterally offset from each other. For example, rack member **310** can include an S-shaped or N-shaped bend that defines the transition between first portion **312** and second portion **314**. In an exemplary embodiment, first portion **312** and second portion **314** both define a longitudinal axis that extends therethrough. The longitudinal axes of first portion **312** and second portion **314** can be substantially parallel to each other and offset in one plane. For example, first portion **312** and second portion **314** can be offset from each other in a plane that is substantially parallel with a bottom or top surface of carriage **46**. In some embodiments, first portion **312** and second portion **314** are substantially parallel to each other and are offset a distance in a plane that is substantially perpendicular to one of or both axes **45**. Axis **45***a* and axis **45***b* can be substantially parallel to each other and a distance therebetween is substantially equal to an overall width of carriage **46**.
(66) Rotation of output gear **318** in a first direction translates one of rack member **310** and rack member **309** in a first direction (e.g., direction **322**) and the other one of rack member **310** and rack member **309** in a second direction (e.g., direction **320**) that is opposite the first direction. Likewise, rotation of output gear **318** in an opposite direction translates rack member **310** and rack member **309** in opposite directions. In some embodiments, rack member **310** and rack member **309** are configured to translate in opposite directions. For example, as rack member **310** translates in direction **322**, rack member **309** translates in direction **320**. Likewise, as rack member **310** translates in direction **320**, rack member **309** translates in direction **322**.
(67) For example, rotation of output gear **318** in direction **324** (e.g., clockwise) can drive rack member **309** to translate in direction **320**, thereby pivoting first grabber arm **44***a* outwards (through the meshed coupling between teeth **308** of member **306** and arm gear **302***a*). Rotation of output gear **318** in direction **324** also drives rack member **310** to translate in direction **322**, thereby pivoting second grabber arm **44***b* outwards (through the meshed coupling between teeth **316** of second portion **314** and arm gear **302***b*). Similarly, rotation of output gear **318** in a direction opposite direction **324** drives rack member **309** to translate in direction **322** (thereby pivoting grabber arm **44***a* inwards) and drives rack member **309** to translate in direction **320** (thereby pivoting second grabber arm **44***b* inwards). In this way, electric motor **354** can be operated to rotate grabber arms **44** inwards (to grasp a refuse container) or to rotate grabber arms **44** outwards (to release a refuse container or spread grabber arms **44** apart).
(68) In some embodiments, rack member **309** defines a longitudinal axis that extends therethrough. The longitudinal axis that extends through rack member **309** can be substantially parallel to one or both of the longitudinal axes of first portion **312** and second portion **314** of rack member **310**. In some embodiments, rack member **309** and rack member **310** are meshingly coupled (e.g., through teeth) on opposite sides of output gear **318**. Rack members **309** and **310** can be rotatably fixed relative to carriage **46**. For example, rack members **309** and **310** can be configured to slidably couple with a track, a recess, a groove, a guide member, etc., of carriage **46**.
(69) Referring now to FIG. **12**, another electric rack and pinion system **400** is shown, according to an exemplary embodiment. Electric rack and pinion system **400** can include an electric linear actuator **450** that drives rack members, bars, beams, rods, elongated members, etc., shown as rack member **406** and rack member **414** to translate, thereby pivoting first grabber arm **44***a* and second grabber arm **44***b* to grasp or release a refuse container.

(70) Electric linear actuator **450** includes an electric motor **454** and an extension member **402**. Extension member **402** can include a lead screw and a nut that is coupled with an output driveshaft of electric motor **454** and transfers the rotational output of electric motor **454** to translatory motion. Electric linear actuator **450** includes an output shaft **404** configured to extend and retract. Output shaft **404** is configured to extend and retract in response to operation (e.g., a direction of rotation) of electric motor **454**. Output shaft **404** can be received within extension member **402** and may slidably couple with an apertures of extension member **402**. In some embodiments, output shaft **404** is coupled with the lead screw of extension member **402** and translates (e.g., extends or retracts) with the lead screw. In some embodiments, extension member **402** is or includes a ball screw. For example, a portion of output shaft **404** can include ball screw threads along a ball screw portion, and extension member **402** can include a ball nut configured to threadingly couple with the ball screw portion of output shaft **404**.
(71) An end portion of output shaft **404** can be removably or fixedly coupled with a connection portion (e.g., a flange, a protrusion, etc.) of rack member **406**. For example, the end portion of output shaft **404** can be removably coupled with the connection portion of rack member **406** with fasteners. In some embodiments, the end portion of output shaft **404** can rotate relative to rack member **406**. In other embodiments, the end portion of output shaft **404** is fixedly coupled with rack member **406**. Output shaft **404** can provide pushing or pulling forces to rack member **406** through the coupling of the end portion of output shaft **404** and the corresponding connection portion of rack member **406**.
(72) Rack member **406** includes teeth **408** configured to mesh with a corresponding arm gear **410***a*. Arm gear **410***a* can be fixedly coupled (e.g., through a keyed connection, a press or interference fit, etc.) with bushing **54***a*. In some embodiments, arm gear **410***a* is fixedly coupled with an exterior surface of bushing **54***a*. Arm gear **410***a* can be rotated about axis **45***a* to pivot or rotate first grabber arm **44***a* about axis **45***a*.
(73) Rack member **406** is coupled with rack member **414** through a structural member, a coupling member, an elongated member, a rod, etc., shown as tie rod **412**. Tie rod **412** can extend therebetween rack member **406** and rack member **414** and couple rack member **406** and rack member **414**. Tie rod **412** can translatably couple rack member **406** and rack member **414** with each other. Tie rod **412** can removably or fixedly couple at either end with corresponding portions of rack member **406** and rack member **414**.
(74) Rack member **414** includes teeth **409** configured to mesh with a gear **416**. Gear **416** is rotatably coupled with carriage **46**. Gear **416** can be translationally fixedly coupled with carriage **46**. Gear **416** can be rotatably coupled with carriage **46** with a fixed or rotatable shaft. For example, gear **416** can be mounted on a shaft (e.g., with a keyed interface, with a press fit, etc.) that is rotatably coupled (e.g., with a bearing) with carriage **46**. In other embodiments, the shaft is fixedly coupled with carriage **46** and gear **416** is rotatably coupled with the shaft (e.g., through a bearing) such that gear **416** can rotate relative to carriage **46**. In either case, gear **416** is rotatable relative to carriage **46** and is translationally fixedly coupled with carriage **46**.
(75) Gear **416** is configured to drive arm gear **410***b*. Arm gear **410***b* can be the same as or similar to arm gear **410***a*. Arm gear **410***b* can be fixedly coupled or press fit with an outer periphery (e.g., an outer surface) of bushing **54***b*. Arm gear **410***b* is configured to rotate relative to carriage **46** to facilitate pivoting/rotation of second grabber arm **44***b* about axis **45***b* relative to carriage **46**. Translatory motion of rack member **414** drives arm gear **410***b* to pivot/rotate about axis **45***b*, thereby rotating/pivoting second grabber arm **44***b*.
(76) In some embodiments, tie rod **412** includes an outer member **417** and an inner member **420**. Inner member **420** is configured to be received within outer member **417**. In some embodiments, inner member **420** is threadingly coupled with an inner surface of outer member **417** and can be extended or retracted relative to outer member **417**. Tie rod **412** can include a locking member, shown as lock nut **418**. Lock nut **418** can be tightened once a desired extension between outer member **417** and inner member **420** is achieved to fixedly couple outer member **417** and inner member **420**.
(77) Lock nut **418** can be loosened and inner member **420** and outer member **417** can be adjusted to a desired extension. In some embodiments, inner member **420** and outer member **417** can be adjusted to increase or decrease an overall length of tie rod **412**. Increasing the overall length of tie rod **412** results in first grabber arm **44***a* and second grabber arm **44***b* being pivoted closer (e.g. into a more-grasped configuration) together. Likewise, decreasing the overall length of tie rod **412** results in first grabber arm **44***a* and second grabber arm **44***b* being pivoted further apart (e.g., into a more-released configuration). In this way, a distance between first grabber arm **44***a* and second grabber arm **44***b* when in the fully grasped configuration can be adjusted (e.g., increased) to facilitate grasping various sized refuse containers.
(78) Translatory motion of rack member **406** results in arm gear **410***a* being rotated about axis **45***a* and arm gear **410***b* being rotated about axis **45***b*, thereby pivoting first grabber arm **44***a* and second grabber arm **44***b*. Rack member **406** and rack member **414** are configured to translate in a same direction (e.g., in direction **322** or direction **320**), thereby causing pivotal/rotatable motion of first grabber arm **44***a* and second grabber arm **44***b* in opposite directions (e.g., inwards to grasp a container or outwards to release a container or outwards to release a container).
(79) For example, electric motor **454** can be driven to rotate in a first direction to extend output shaft **404**, thereby translating rack member **406** and rack member **414** in direction **322**. As rack member **406** and rack member **414** are translated in direction **322**, arm gear **410***a* is driven to rotate about axis **45***a* in a counter-clockwise direction, while arm gear **410***b* is driven to rotate about axis **45***b* in a clockwise direction. Likewise, electric motor **454** can be driven to rotate in a second direction to retract output shaft **404**, thereby translating rack member **406** and rack member **414** in direction **320** (opposite direction **322**). As rack member **406** and rack member **414** are translated in direction **320**, arm gear **410***a* is driven to rotate about axis **45***a* in a clockwise direction about axis and arm gear **410***b* is driven to rotate about axis **45***b* in a counter-clockwise direction. In this way, electric motor **454** can be operated to pivot/rotate first grabber arm **44***a* and second grabber arm **44***b* to grasp and release refuse containers.
(80) Referring still to FIG. **12**, rack member **406** and rack member **414** can each include a channel, a slot, a recess, etc., shown as track **422** and track **424**, respectively. Particularly, rack member **406** includes track **422** and rack member **414** includes track **424**. Track **424** can have a square shape, a trapezoidal shape, a curved shape, an irregular shape, etc. Track **424** and track **422** can be the same as or similar to each other. Track **424** and track **422** are configured to slidably couple with a corresponding surface, protrusion, grooves, member, etc., of carriage **46**. Track **422** and track **424** facilitate translator (e.g., translational) motion of rack member **406** and rack member **414**, respectively. Additionally, track **422** and track **424** facilitate alignment of rack member **406** and rack member **414**, respectively.

(81) Referring particularly to FIG. **13**, an electric lead screw system **500** is shown, according to an exemplary embodiment. Electric lead screw system **500** can be implemented with electric rack and pinion system **400**, or electric rack and pinion system **300**. Electric lead screw system **500** can be used to transfer rotational motion from an electric motor to translator motion.
(82) Electric lead screw system **500** includes an electric motor **502**. In some embodiments, electric motor **502** is the same as or similar to electric motor **454**, electric motor **354**, and/or electric motors **154**. Electric motor **502** is configured to output rotational kinetic energy through a driveshaft **504**. In some embodiments, electric motor **502** includes an internal gearbox that increases torque output through driveshaft **504**. Driveshaft **504** defines a longitudinal axis **522** and is configured to rotate about longitudinal axis **522**. Driveshaft **504** includes an output gear **506**. Output gear **506** can be a spur gear, a helical gear, etc., or any other type of gear. Output gear **506** is fixedly and rotatably coupled with driveshaft **504**. In some embodiments, output gear **506** is press fit onto driveshaft **504**. In other embodiments, output gear **506** is fixedly and rotatably coupled with driveshaft **504** with a key.
(83) Output gear **506** is configured to mesh with a drive gear **508**. Output gear **506** can transfer the rotational kinetic energy output to an elongated member **514** with which drive gear **508** is fixedly and rotatably coupled. In an exemplary embodiment, elongated member **514** includes threads along portions at opposite ends. The threaded end portions of elongated member **514** are configured to be received within and threadingly couple with corresponding apertures, inner volumes, receiving portions, threads, etc., of outer members **518** and **516**. In some embodiments, outer members **518** and **516** are outer threaded portions of elongated member **514**. Outer members **518** and **516** can each receive the threaded end portions of elongated member **514** within a correspondingly shaped and threaded inner volume, inner bore, aperture, etc. In other embodiments, outer members **518** and **516** are configured to be received within and threadingly couple with a correspondingly threaded aperture, bore, hole, etc., of elongated members **510** and **512**, respectively. The threaded end portions of elongated member **514** can have oppositely oriented threads. For example, one of the threaded end portions may have right-hand oriented threads, while the other threaded end portion may have left-hand oriented threads. In some embodiments, the threaded end portions of elongated member **514** have similarly oriented threads (e.g., both are left-hand oriented or both are right-hand oriented) and outer members **518** and **516** are configured to receive and threadingly couple with the threaded end portions of elongated member **514**.
(84) As electric motor **502** is driven, output gear **506** drives drive gear **508**, thereby driving elongated member **514** to rotate about longitudinal axis **520**. Longitudinal axis **520** extends therethrough elongated member **514**. Outer members **518** and **516** can be rotatably stationary relative to elongated member **514** such that as elongated member **514** rotates, outer members **518** and **516** do not rotate, thereby resulting in relative rotation between elongated member **514** and outer members **518** and **516**. The threaded coupling between elongated member **514** and outer members **518** and **516** and the relative rotation between elongated member **514** and outer members **518** and **516** results in outer members **518** and **516** being driven (e.g., translated) apart or translated closer together. Likewise, if outer members **518** and **516** are outer threaded portions of elongated member **514**, rotation of elongated member drive elongated members **510** and **512** to translate along elongated member **514** (e.g., along an entire length of the threads of outer members **518** and **516**).
(85) Outer members **518** and **516** can include elongated members, bars, beams, structural components, etc., shown as elongated members **510** and **512**. Elongated members **510** and **512** can be fixedly coupled with outer members **518** and **516**. In some embodiments, elongated members **510** and **512** are threadingly coupled with outer members **518** and **516** and are configured to translate along outer members **518** and **516** of elongated member **514** as elongated member **514** rotates. In some embodiments, elongated members **510** and **512** are mounted to outer members **518** and **516**. Elongated members **510** and **512** can be any structural component, translatory component, etc., of any of electric rack and pinion system **400** or electric rack and pinion system **300**. Elongated members **510** and **512** are shown spaced a distance **524** apart. Operating electric motor **502** in a first direction results in distance **524** increasing (as elongated members **510** and **512** are driven apart, while operating electric motor **502** in a second direction results in distance **524** decreasing (as elongated members **510** and **512** are driven closer together). For example, elongated members **510** and **512** can each be rack and pinion members, each configured to drive one or more gears to pivot first grabber arm **44***a* and second grabber arm **44***b*. In some embodiments, outer ends of outer member **518** and outer member **516** are pivotally coupled with carriage **46** and one of grabber arms **44** to drive grabber arms **44** to pivot.
(86) Referring now to FIG. **14**, an electric system **600** for operating grabber assembly **42** includes electric linear actuators **602**. Electric linear actuators **602** are each configured to pivot/rotate grabber arms **44** about axes **45**. Specifically, electric linear actuator **602***a* is configured to pivot/rotate first grabber arm **44***a* about axis **45***a*, while electric linear actuator **602***b* is configured to pivot/rotate second grabber arm **44***b* about axis **45***b*.
(87) Electric linear actuators **602** each include electric motors **604**. Electric motors **604** can be the same as or similar to any of the other electric motors described herein. In some embodiments, any of electric motors **604** (or any other electric motors described herein) include gear boxes to increase or decrease torque output. Any of electric motors **604** (or any other electric motors described herein) can also include a brake. Electric motors **604** can be operated by a controller or a control system.
(88) Electric linear actuators **602** include an extendable member **606**. Extendable member **606** is configured to be driven by electric motor **604** to expand or retract. Extendable member **606** can include telescoping members (e.g., an inner member and an outer member that receives the inner member therewithin). The inner member of extendable members **606** can be driven by electric motors **604** translate (e.g., extend or retract) relative to the outer member, thereby pivoting or rotating a corresponding grabber arm **44** about the corresponding axis **45**.
(89) For example, electric linear actuator **602***a* can be pivotally or rotatably coupled with first grabber arm **44***a* at a first or proximate end **608***a* and pivotally or rotatably coupled with carriage **46** at a second or distal end **610***a*. Likewise, electric linear actuator **602***b* can be pivotally or rotatably coupled with second grabber arm **44***b* at a first or proximate end **608***b* and pivotally or rotatably coupled with carriage **46** at a second or distal end **610***b*. Electric linear actuator **602***a* and electric linear actuator **602***b* can be independently operated to pivot first grabber arm **44***a* and second grabber arm **44***b* about axis **45***a* and axis **45***b*, respectively. In some embodiments, electric linear actuator **602***a* and electric linear actuator **602***b* are operated by a control system simultaneously/concurrently with each other to cause grabber assembly **42** to grasp or release a refuse container/receptacle.
(90) Electric linear actuator **602***a* can be pivotally or rotatably coupled with first grabber arm **44***a* at a position such that extension of extendable member **606***a* (e.g., translation of the inner member relative to the outer member that increases an overall length of

extendable member **606***a*) rotates first grabber arm **44***a* about axis **45***a* in a counter-clockwise direction. Similarly, retraction of extendable member **606***a* (e.g., translation of the inner member relative to the outer member that decreases an overall length of extendable member **606***a*) can rotate first grabber arm **44***a* about axis **45***a* in a clockwise direction. In some embodiments, electric linear actuator **602***a* is coupled with first grabber arm **44***a* at proximate end **608***a* such that extension of extendable member **606***a* produces a torque about axis **45***a* in a counter-clockwise direction.

(91) Likewise, electric linear actuator **602***b* can be pivotally or rotatably coupled with second grabber arm **44***b* at a position such that extension of extendable member **606***b* (e.g., translation of the inner member relative to the outer member that increases an overall length of extendable member **606***b*) rotates second grabber arm **44***b* about axis **45***b* in a clockwise direction. Likewise, retraction/compression of electric linear actuator **602***b* can cause second grabber arm **44***b* to rotate about axis **45***b* in a counter-clockwise direction. Electric linear actuator **602***b* can extend or retract to pivot/rotate second grabber arm **44***b* in either direction.

(92) Advantageously, electric system **600**, electric lead screw system **500**, electric rack and pinion system **400**, electric rack and pinion system **300**, and gearing system **200** are fully electric systems or are configured to be driven by electric motors, thereby facilitating a fully electric grabber assembly. While the various electric systems described herein are shown implemented with grabber assembly **42**, any of the electric systems, the electric rack and pinion systems, the gearing systems, electric linear actuators, electric motors, etc., or components thereof can be used with various grabber assemblies. Advantageously, a fully-electric grabber reduces the need for a hydraulic system, is more environmentally friendly, and facilitates a more robust grabber.

(93) It should be understood that any of the electric motors, electric linear actuators, electric devices, etc., can receive electrical energy/power from a battery system including one or more battery devices or any other energy storage devices. Similarly, any of the electric motors, electric linear actuators, or electrical devices described herein can be operated by a controller or a control system. The controller can include a processing circuit, memory, a processor, computer readable medium, etc., and may store instructions for operating any of the functions of a grabber assembly. The controller can generate control signals and provide the control signals to any of the electrical devices (e.g., the electric motors) described herein.

(94) It should also be noted that any of the electric motors, electric linear actuators, etc., can include a brake that can lock or facilitate restricting rotational output from an output driveshaft of any of the electric motors. For example, any of the electric motors can include a drum brake configured to activate and provide a frictional force to the electric motor driveshaft to facilitate preventing rotation of the driveshaft thereof. The brake can be activated using mechanical systems, or an electrical system. For example, the brake may be an electrically activated drum brake, a mechanical brake, an electrical brake, etc. The brake can be configured to decrease output speed of the driveshaft of the electric motor or to facilitate locking a current angular position of the driveshaft of the electric motor. The brake can be operated by the same controller or control system that operates the electric motors and electric linear actuators, or can be operated by a separate control system and/or a separate controller. Additionally, any of the electric motors or linear electric actuators described herein can include appropriate gearboxes to increase or decrease output torque.

(95) It should also be noted that any of the electrical motors, electrical actuators, or any other electrical movers can include any number of sensors configured to measure and monitor an angular position or a degree of extension. In some embodiments, the sensors are a component of the electric motors or the electric linear actuators and provide feedback signals to the controller. The controller can monitor the sensor signals to identify an angular position or a degree of extension of the electric motors or the electric linear actuators, respectively. The controller can use the sensor signal to determine a current angular orientation of grabber arms **44**. In some embodiments, angular orientation of grabber arms **44** is measured directly (e.g., with a rotary potentiometer).

(96) Referring particularly to FIGS. **20-22**, grabber assembly **42** is shown according to another embodiment. Grabber assembly **42** includes first grabber arm **44***a*, second grabber arm **44***b*, carriage **46**, and connecting member **26**. Grabber assembly **42** of FIGS. **20-22** can be the same as or similar to any of the embodiments of grabber assembly **42** as described in greater detail above with reference to FIGS. **1-19**. First grabber arm **44***a* is configured to rotatably or pivotally couple with carriage **46** such that first grabber arm **44***a* can rotate or pivot relative to carriage **46** about axis **45***a*. Likewise, second grabber arm **44***b* is configured to rotatably or pivotally couple with carriage **46** such that second grabber arm **44***b* can rotate or pivot relative to carriage **46** about axis **45***b*.

(97) Grabber assembly **42** can include an electric gripping motor system **1000** and an electric climb system **2000**. Electric gripping motor system **1000** includes an electric motor, shown as gripping motor **1002** that is configured to operate to drive first grabber arm **44***a* and second grabber arm **44***b* to pivot or rotate relative to carriage **46** about axis **45***a* and axis **45***b*, respectively.

(98) Connecting member **26** can include a first lateral member **92***a* and a second lateral member **92***b* that are spaced a distance apart. First lateral member **92***a* and second lateral member **92***b* are substantially parallel with each other and spaced apart so that first lateral member **92***a* may be positioned outside of track **20** at a first lateral side of track **20**, and so that second lateral member **92***b* may be positioned outside of track **20** at a second lateral side of track **20**. First lateral member **92***a* includes a first roller **94***a* and second lateral member **92***b* includes a second roller **94***b*. First roller **94***a* and second roller **94***b* can be rotatably coupled with first lateral member **92***a* and second lateral member **92***b*, respectively, so that as grabber assembly **42** ascends or descends along track **20**, first roller **94***a* and second roller **94***b* rotate. In some embodiments, first roller **94***a* and second roller **94***b* are fixedly coupled with first lateral member **92***a* and second lateral member **92***b*, respectively. First roller **94***a* can be received within a corresponding groove, track, recess, etc., of track **20** (e.g., recess **96***a* shown in FIG. **23**) and second roller **94***b* can be received within a corresponding groove, track, recess, etc., of track **20** on an opposite lateral side of track **20** (e.g., recess **96***b* as shown in FIG. **23**). First roller **94***a* and second roller **94***b* can be positioned at an upper end, or an upper portion of connecting member **26**. Connecting member **26** can also include a second pair of rollers **95** (e.g., first roller **95***a* and second roller **95***b*) that are positioned at a lower end or lower portion of connecting member **26** and are similarly configured to engage, be received within, etc., the tracks, grooves, recesses, etc., of track **20**.

(99) Referring particularly to FIGS. **21** and **22**, electric climb system **2000** includes an electric motor, shown as climb motor **2002**, an output gear **2004**, and a driven gear **2006**. Output gear **2004** can be a spur gear, a helical gear, etc., that is rotatably fixedly coupled with an output driveshaft of climb motor **2002**. Climb motor **2002** can operate to provide rotational kinetic energy to output gear **2004**. Climb motor **2002** may drive output gear **2004** which meshes, engages, transfers rotational kinetic energy or

torque, etc., to driven gear **2006**. Output gear **2004** and driven gear **2006** may be configured to rotate about axes that are parallel with each other and spatially offset. Output gear **2004** may have a greater number of teeth and a greater diameter than driven gear **2006** so that an angular speed of driven gear **2006** is greater than an angular speed of output gear **2004**. In this way, driven gear **2006** and output gear **2004** may form a gear train, a gear assembly, a gear sub-assembly, a drive train, etc., to transfer rotational kinetic energy or torque from climb motor **2002** to a pinion **2008**.

(100) Referring particularly to FIGS. **21-23**, electric climb system **2000** includes pinion **2008** that is positioned between first lateral member **92***a* and second lateral member **92***b*. Pinion **2008** can be a spur gear, a cogged drum, a drive drum, a gear, a pinion, a rotational engagement member, engagement device, a rotatable climb member, etc., including one or more radial protrusions that are configured to engage, abut, contact, etc., corresponding rungs **98** of track **20**. Rungs **98** of track **20** may extend between opposite frame members of track **20** that define recess **96***a* and recess **96***b*. In some embodiments, track **20** functions as a rack and pinion **2008** is driven by climb motor **2002** to drive grabber assembly **42** to ascend or descend along track **20**.

(101) Referring still to FIGS. **21-23**, pinion **2008** can be positioned between first lateral member **92***a* and second lateral member **92***b*. For example, pinion **2008** can be rotatably fixedly coupled with an output shaft **2012** that extends between first lateral member **92***a* and second lateral member **92***b* of connecting member **26**. Output shaft **2012** may be rotatably coupled at opposite ends with first lateral member **92***a* and second lateral member **92***b* through bearings **2010**. In some embodiments, a first end of output shaft **2012** is rotatably fixedly coupled with driven gear **2006** so that rotational kinetic energy or torque is transferred from climb motor **2002**, through an output driveshaft of climb motor **2002**, through output gear **2004** and driven gear **2006**, and to output shaft **2012** and pinion **2008**. In this way, climb motor **2002** can rotate in a first direction so that pinion **2008** rotates to drive grabber assembly **42** to ascend or climb track **20**, and in a second direction so that pinion **2008** rotates to drive grabber assembly **42** to descend track **20** (or so that grabber assembly descends track **20** at a controlled rate).

(102) Referring particularly to FIG. **22**, electric gripping motor system **1000** includes gripping motor **1002**, output driveshaft **1004**, output gear **1008**, driven gear **1012**, and an electric brake **1010**. Electric gripping motor system **1000** also includes a medial or intermediate shaft assembly **1024** that is configured to deliver or transfer power provided by gripping motor **1002** to first grabber arm **44***a* and second grabber arm **44***b* to drive first grabber arm **44***a* and second grabber arm **44***b* to pivot about first axis **45***a* and second axis **45***b*, respectively (e.g., to grasp and/or release a container or refuse bin).

(103) Gripping motor **1002** is configured to operate to generate or provide rotational kinetic energy or torque that is transferred through output driveshaft **1004**. Output driveshaft **1004** may be rotatably coupled with connecting member **26** (e.g., first lateral member **92***a*) through a bearing **1006** so that output driveshaft **1004** is supported by connecting member **26** and can rotate relative to connecting member **26**. In some embodiments, output driveshaft **1004** is configured to be selectively engaged by electric brake **1010**. For example, electric brake **1010** can receive an electrical current or electrical power from a battery, power storage device, etc., of refuse vehicle **10** and operate to engage, lock, interface with, etc., output driveshaft **1004** so that output driveshaft **1004** is locked at a current angular position or to restrict or prevent rotation of output driveshaft **1004**. In some embodiments, electric brake **1010** is transitionable between a first position (e.g., an unlocked position) so that rotation of output driveshaft **1004** is not limited (e.g., output driveshaft **1004** is freely driven by gripping motor **1002**) and a second position (e.g., a locked position) so that rotation of output driveshaft **1004** is limited, prevented, restricted, etc. (e.g., so that output driveshaft **1004** is limited in its rotation or maintained at a current angular position or maintained within a specific angular range). Electric brake **1010** can transition between the first position and the second position in response to receiving a signal from the controller of refuse vehicle **10**.

(104) Output gear **1008** engages, meshes with, etc., driven gear **1012** and transfers rotational kinetic energy or torque to driven gear **1012**. Output gear **1008** and driven gear **1012** can be spur gears, helical gears, etc., or any other types of gears. Driven gear **1012** may be rotatably fixedly coupled with a first shaft **1014***a*. In some embodiments, first shaft **1014***a* is rotatably coupled with connecting member **26** (e.g., first lateral member **92***a*) through a bearing **1016** so that first shaft **1014***a* can rotate relative to connecting member **26**. First shaft **1014***a* includes a first end and a second end. The first end of first shaft **1014***a* can include screw threads, worm threads, a worm drive, etc., shown as first worm **1018***a*. First worm **1018***a* is configured to engage, mesh with, etc., a corresponding first worm gear **1020***a* that is rotatably fixedly coupled with first grabber arm shaft **1022***a*. First grabber arm shaft **1022***a* can be the same as or similar to first bushing **54***a*. In some embodiments, first grabber arm shaft **1022***a* is the same as or similar to adapter assembly pin **60***a*.

(105) First grabber arm shaft **1022***a* may define axis **45***a*. In some embodiments, first grabber arm shaft **1022***a* is fixedly coupled at opposite ends with a first control arm **90***a* and a second control arm **90***b* of first grabber arm **44***a* (shown in FIGS. **20-22**). First grabber arm shaft **1022***a* can be rotatably supported within a first housing or a first structural member **38***a* that is fixedly coupled with carriage **46** and/or connecting member **26**. In some embodiments, first grabber arm shaft **1022***a* is rotatably coupled with first structural member **38***a* through one or more bearings.

(106) In this way, gripping motor **1002** may be operated to drive first grabber arm **44***a* to rotate about axis **45***a* to grasp, grip, or otherwise removably couple with a container. Gripping motor **1002** outputs rotational kinetic energy or torque through output driveshaft **1004** which is transferred to output gear **1008**. Output gear **1008** drives driven gear **1012** which is rotatably fixedly coupled with first shaft **1014***a* so that rotational kinetic energy is transferred through driven gear **1012** to first shaft **1014***a*. First shaft **1014***a* rotates to drive first worm gear **1020***a*, first grabber arm shaft **1022***a*, first control arm **90***a*, second control arm **90***b*, and first grabber arm **44***a* to rotate about axis **45***a* (e.g., to grasp and release a refuse container). Gripping motor **1002** can operate to drive output driveshaft **1004** in a first direction to drive first grabber arm **44***a* to rotate about axis **45***a* in a first direction (e.g., inwards, counter-clockwise, etc.) to grasp a container and can operate to drive output driveshaft **1004** in a second direction to drive first grabber arm **44***a* to rotate about axis **45***a* in a second or opposite direction (e.g., outwards, clockwise, etc.) to release a container.

(107) In some embodiments, first shaft **1014***a* extends in a direction that is substantially orthogonal or perpendicular to axis **45***a*. First shaft **1014***a* can rotatably couple (e.g., fixedly) with intermediate shaft assembly **1024** through a first universal joint **1026***a*. Intermediate shaft assembly **1024** rotatably couples with first shaft **1014***a* through first universal joint **1026***a* at a first end of intermediate shaft assembly **1024**, and rotatably couples with a second shaft **1014***b* through a second universal joint **1026***b* at a second, opposite, or distal end of intermediate shaft assembly **1024**. Second universal joint **1026***b* can be the same as or similar to

first universal joint **1026***a* and/or may be mirrored so that whatever is said of first universal joint **1026***a* may be said of second universal joint **1026***b*. Second shaft **1014***b* can be the same as or similar to first shaft **1014***a* so that whatever is said of first shaft **1014***a* may be said of second shaft **1014***b* and vice versa.

(108) Second shaft **1014***b* includes a second worm **1018***b* that is the same as or similar to first worm **1018***a*. In some embodiments, second worm **1018***b* has a thread direction that is opposite a thread direction of first worm **1018***a*. Second worm **1018***b* is configured to engage, mesh with, etc., a second worm gear **1020***b*. Second worm gear **1020***b* can be the same as or similar to first worm gear **1020***a*. Second worm gear **1020***b* receives rotational kinetic energy or torque from second worm **1018***b* so that second worm gear **1020***b* rotates about axis **45***b*. Second worm gear **1020***b* is fixedly coupled with a second grabber arm shaft **1022***b* that is fixedly coupled with a first control arm **90***a* and second control arm **90***b* of second grabber arm **44***b* (e.g., at opposite ends of second grabber arm shaft **1022***b*). In this way, a single gripping motor **1002** can be used to drive both first grabber arm **44***a* and second grabber arm **44***b* to rotate about axis **45***a* and axis **45***b*, respectively. Specifically, intermediate shaft assembly **1024** facilitates providing rotational kinetic energy or torque for both first grabber arm **44***a* and second grabber arm **44***b*.

(109) In some embodiments, first universal joint **1026***a* and second universal joint **1026***b* are optional. For example, intermediate shaft assembly **1024** may extend direction between first worm **1018***a* and second worm **1018***b* without first universal joint **1026***a* and second universal joint **1026***b*. In some embodiments, first worm **1018***a* and second worm **1018***b* are formed directly or integrally formed with opposite ends of intermediate shaft assembly **1024**. First universal joint **1026***a* and second universal joint **1026***b* can facilitate accounting for any mis-alignment between first worm **1018***a* and second worm **1018***b*.

(110) In other embodiments, electric gripping motor system **1000** includes a first gripping motor **1002** and a second gripping motor **1002**. The first gripping motor **1002** can be configured to drive first grabber arm **44***a* to rotate about axis **45***a* as shown in FIG. **22** without intermediate shaft assembly **1024**. The second gripping motor can be configured to independently drive second grabber arm **44***b* to rotate about axis **45***b* using a gear train similar to output driveshaft **1004**, output gear **1008**, driven gear **1012**, first shaft **1014***a*, first worm **1018***a*, and first worm gear **1022***a*. In this way, a similar but mirrored gear train can be provided on an opposite side of grabber assembly **42** with its own gripping motor **1002** for independently driving second grabber arm **44***b*. The first gripping motor and second gripping motor can operate cooperatively to independently drive each of the first grabber arm **44***a* and the second grabber arm **44***b* to grasp and release refuse containers.

(111) In some embodiments, first worm **1018***a* and second worm **1018***b* function to provide locking functionality or to reduce a likelihood that grabber arms **44** back-drive. For example, first worm **1018***a* and second worm **1018***b* may transfer rotational kinetic energy to first worm gear **1020***a* and second worm gear **1020***b*, respectively, to pivot grabber arms **44** about their respective axes **45**, and prevent, restrict, limit, or reduce the likelihood that first worm **1018***a* and second worm **1018***b* are back driven by rotation of grabber arms **44** about their respective axes **45**. In some embodiments, due to the anti-back driving characteristic of the engagement between worms **1018** and worm gears **1020**, brake **1010** is optional.

Configuration of Exemplary Embodiments

(112) As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(113) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(114) The term “coupled,” as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. Such members may be coupled mechanically, electrically, and/or fluidly.

(115) The term “or,” as used herein, is used in its inclusive sense (and not in its exclusive sense) so that when used to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is understood to convey that an element may be either X, Y, Z; X and Y; X and Z; Y and Z; or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

(116) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” etc.) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(117) The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The

memory (e.g., memory, memory unit, storage device, etc.) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit and/or the processor) the one or more processes described herein.

(118) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(119) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(120) It is important to note that the construction and arrangement of the fire suppression system as shown in the various exemplary embodiments is illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. A fully-electric grabber assembly comprising: a first grabber arm; a second grabber arm; an electric motor; and a plurality of gears including: a first gear coupled with the electric motor, the first gear configured to rotate about an axis radially offset from a center thereof; an intermediate gear coupled with the first gear; an arm gear coupled with the first grabber arm and the intermediate gear to facilitate pivoting the first grabber arm; and at least one of (a) a first linkage extending between the center of the first gear and a center of the intermediate gear or (b) a second linkage extending between the center of the intermediate gear and a center of the arm gear.
2. The fully-electric grabber assembly of claim 1, wherein the intermediate gear is positioned to be driven by the first gear, and the arm gear is positioned to be driven by the intermediate gear.
3. The fully-electric grabber assembly of claim 1, wherein the plurality of gears form an eccentric gear train.
4. The fully-electric grabber assembly of claim 1, wherein the plurality of gears are configured to rotate about axes that are substantially parallel with an axis about which the first grabber arm rotates.
5. The fully-electric grabber assembly of claim 1, wherein the plurality of gears are a first plurality of gears and the electric motor is a first electric motor, further comprising a second electric motor and a second plurality of gears configured to drive the second grabber arm to rotate independently of the first grabber arm.
6. The fully-electric grabber assembly of claim 1, wherein at least one of (a) the first gear and the intermediate gear have a same number of teeth or (b) the arm gear comprises teeth along only a portion of an outer periphery thereof.
7. The fully-electric grabber assembly of claim 1, further comprising a carriage, wherein the first grabber arm and the second grabber arm are rotatably coupled with the carriage.
8. A fully-electric grabber assembly for a refuse vehicle, the fully-electric grabber assembly comprising: a carriage; a first grabber arm rotatably coupled with the carriage; a second grabber arm rotatably coupled with the carriage; an electric climb system configured to move the fully-electric grabber assembly relative to the refuse vehicle; and an electric gripping system including: an electric motor; and a gear train coupling the electric motor to the first grabber arm to facilitate pivoting the first grabber arm, the gear train including: a first gear rotatably coupled with the carriage about an axis radially offset from a center of the first gear, the first gear driven by the electric motor; and a pair of second gears coupling the first gear to the first grabber arm; and at least one of (a) a linkage extending between the center of the first gear and a center of a second gear of the pair of second gears or (b) a linkage extending between centers of the pair of second gears.
9. The fully-electric grabber assembly of claim 8, wherein the gear train is a first gear train and the electric motor is a first electric motor, wherein the electric gripping system includes a second electric motor and a second gear train coupling the second electric

motor to the second grabber arm to facilitate pivoting the second grabber arm independently of operation of the first grabber arm.
10. The fully-electric grabber assembly of claim 8, wherein the first gear and at least one of the pair of second gears have a same number of teeth.
11. The fully-electric grabber assembly of claim 8, wherein at least one of the pair of second gears has teeth along only a portion of an outer periphery thereof.
12. The fully-electric grabber assembly of claim 8, wherein the gear train is positioned at least partly within an inner volume of the carriage.
13. A grabber assembly comprising: a grabber arm; an actuator; a first gear driven by the actuator, the first gear rotatable about an axis radially offset from a center thereof; an intermediate gear coupled with the first gear; an arm gear coupled with the grabber arm and the intermediate gear to facilitate pivoting the grabber arm; and at least one of (a) a first linkage extending between the center of the first gear and a center of the intermediate gear or (b) a second linkage extending between the center of the intermediate gear and a center of the arm gear.
14. The grabber assembly of claim 13, wherein the actuator includes an electric motor.
15. The grabber assembly of claim 13, wherein the first gear and the intermediate gear have a same number of teeth.
16. The grabber assembly of claim 13, wherein the arm gear comprises teeth along only a portion of an outer periphery.